Mach-Zehnder interferometer design

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Abstract: This work presents a detailed study on the design and simulation of a Mach–Zehnder interferometer (MZI) using Lumerical, combining analytical modeling with numerical simulations to explore device performance. Systematic parameter sweeps were conducted to examine how variations in structural parameters affect the interferometer's behavior, particularly its spectral response and Free Spectral Range (FSR). The results are compared with theoretical predictions to validate the models and provide deeper insight into the design principles of integrated photonic devices.

1. Introduction

The Mach–Zehnder interferometer (MZI) is a cornerstone device in integrated photonics, widely used for measuring relative phase shifts between two coherent light beams derived from a common source. By splitting light into two separate paths and recombining them, the MZI produces interference patterns that are highly sensitive to variations in optical path length, wavelength, or refractive index.

Owing to its versatility, the MZI forms the foundation for numerous photonic components, including electro-optic and thermo-optic modulators, tunable filters, sensors, and wavelength multiplexers. Its inherent ability to precisely control phase differences makes it an essential tool in applications ranging from high-speed communications to optical sensing and signal processing.

In this work, we present the design and simulation of a silicon-based Mach–Zehnder interferometer, combining mathematical modeling with numerical simulations performed in Lumerical INTERCONNECT and MODE Solutions. Our analysis focuses on understanding how variations in interferometer branch lengths influence its spectral behavior, with particular emphasis on the Free Spectral Range (FSR). The study aims to bridge theoretical predictions with numerical results, offering valuable insights for optimizing MZI performance in integrated photonic systems.

2. Theory

We begin by considering the simplest design of a Mach–Zehnder interferometer (MZI). The device consists of two connected Y-branch waveguides, allowing the splitting and recombination of an input optical signal. Let us denote the input electric field as E_i and the input intensity as I_i.

We assume the electric field propagates as a plane wave, described by:

$$E(z,t) = E_0 e^{i(\omega t - \beta z)}, \tag{1}$$

where ω is the angular frequency of the light, β is the propagation constant of the waveguide, z is the propagation coordinate, and t is time.

At the bifurcation point, the input field splits equally into the two branches, such that each branch carries half of the input intensity. The electric field in each branch can be expressed as:

$$E_{o1} = E_1 e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1}, \tag{2a}$$

$$E_{o2} = E_2 e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}. \tag{2b}$$

Here, α_i represents the optical loss coefficient of each branch, and L_i denotes the respective branch lengths.

When the two branches recombine into a single waveguide, the resulting output electric field is:

$$E_o = \frac{1}{\sqrt{2}} (E_{o1} + E_{o2}) = \frac{E_i}{2} \left(e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right). \tag{3}$$

The corresponding output intensity is then:

$$I_o = \frac{I_i}{4} \left| e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \right|^2. \tag{4}$$

In this work, we consider negligible optical losses ($\alpha_i \approx 0$) and an imbalanced interferometer, i.e., $L_1 \neq L_2$. Under these assumptions, the output intensity simplifies to:

$$I_o = \frac{I_i}{2} \left[1 + \cos(\beta \Delta L) \right], \tag{5}$$

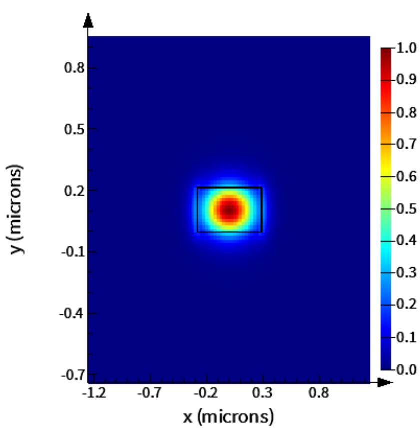
where $\Delta L = L_2 - L_1$ is the path length difference between the two branches.

This fundamental relation describes the interference condition in an MZI and illustrates how variations in ΔL or β directly modulate the output intensity, forming the basis for applications in sensing, modulation, and wavelength filtering.

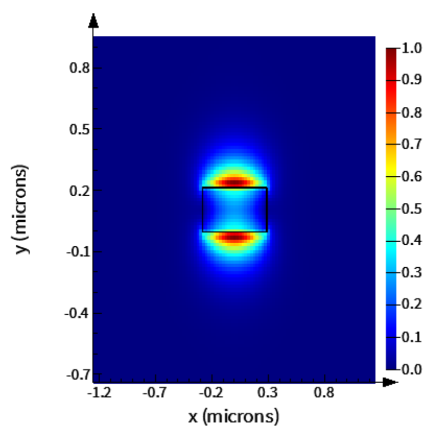
3. Modeling and Simulations

In this design, we consider a silicon (Si) waveguide of width 570 nm and thickness 220 nm, surrounded by a silicon dioxide (SiO_2) cladding.

The transverse electric (TE) and transverse magnetic (TM) mode profiles at $\lambda = 1550$ nm are shown in Figures 1a and 1b, respectively.



(a) TE-mode profile of the waveguide.



(b) TM-mode profile of the waveguide.

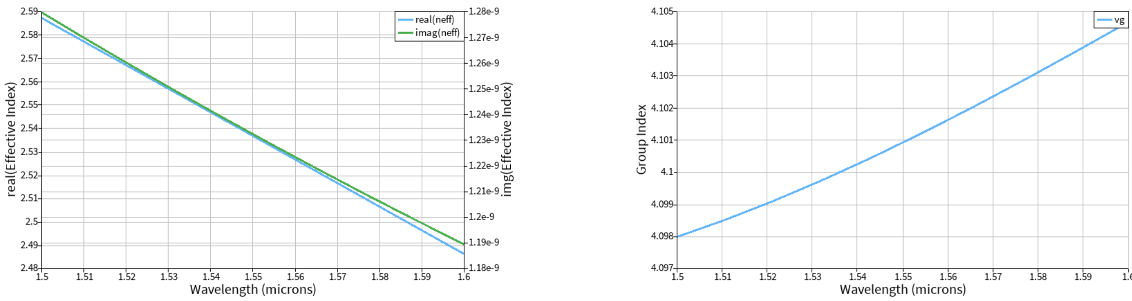
Fig. 1: Mode profiles of the waveguide at $\lambda = 1550$ nm.

For this study, we focus on the TE-mode due to its favorable confinement and propagation characteristics.

The effective index n_{eff} and group index n_g of the waveguide were calculated across a wavelength range. Figures 2a and 2b show their respective dependencies on wavelength.

The effective index dependency on wavelength can be approximated by a polynomial expression:

$$n_{\text{eff}}(\lambda) = 2.537 - 1.009 (\lambda - \lambda_0) - 0.022 (\lambda - \lambda_0)^2, \tag{6}$$



(a) Effective index n_{eff}.

(b) Group index n_g.

Fig. 2: Effective and group indices of the waveguide as a function of wavelength.

where λ_0 is a reference wavelength.

The transfer function of the MZI is given by:

$$T_{\text{MZI}}(\lambda) = \frac{1}{2}\left[1 + \cos\left(\beta \Delta L\right)\right], \tag{7}$$

where $\Delta L = L_2 - L_1$ is the path length difference between the two arms of the interferometer.

The free spectral range (FSR) of the MZI is related to the group index as:

$$\text{FSR} = \frac{\lambda^2}{n_g \, \Delta L}. \tag{8}$$

The schematic of the simulated circuit is shown in Figure 3.

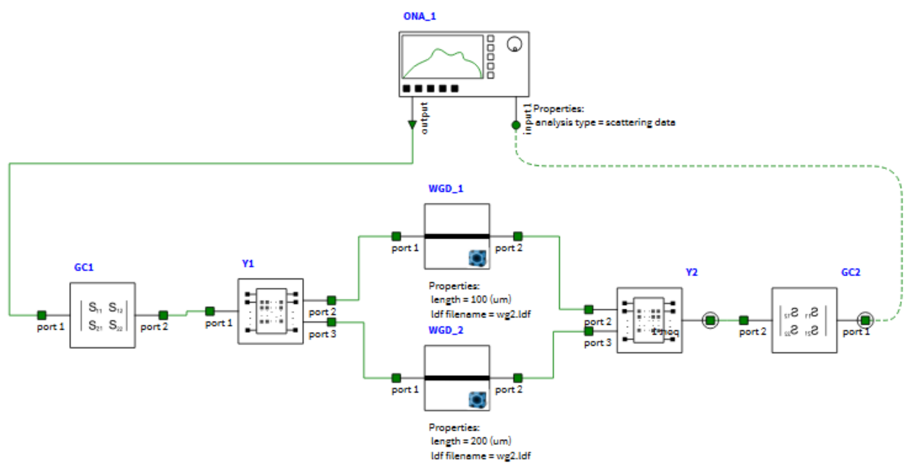


Fig. 3: Schematic diagram of the simulated Mach–Zehnder interferometer.

After simulation, the transmission spectrum of the MZI with and without grating couplers was obtained, as shown in Figure 4. This spectrum illustrates the free spectral range and interference pattern.

Table I summarizes a parameter sweep for different branch lengths, showing the relationship between ΔL and the corresponding FSR.

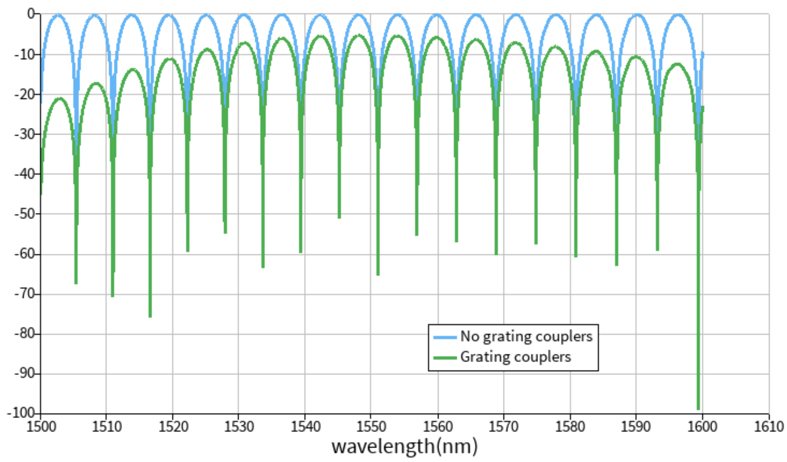


Fig. 4: Transmission spectrum of the MZI for $L_1 = 100$ nm and $L_2 = 200$ nm.

TABLE I: Relationship between branch lengths and free spectral range (FSR).

L_1 (μm)	L_2 (μm)	ΔL (μm)	FSR (nm)
100	50	-50	11.52
100	75	-25	20.17
100	100	0	0
100	150	+50	11.30
100	200	+100	5.75

4. Layout

In this section, the physical layouts of the Mach–Zehnder interferometers (MZIs) designed in this study are presented. A total of seven interferometers were created, each with a different path length difference ΔL, in order to analyze how this parameter influences the spectral response and free spectral range (FSR).

The layouts were designed using the KLayout software, which allows precise control over waveguide geometries, bends, and coupling regions. Each MZI structure includes input and output grating couplers for efficient light coupling and Y-branch splitters for optical power division and recombination.

Figure 5 shows the top-view layout containing the seven MZI devices implemented on the same photonic chip.

5. Comparison with Experimental Results

After the fabrication of the Mach–Zehnder interferometers on a silicon-on-insulator (SOI) platform, the devices were characterized using standard optical measurement techniques. The experimental transmission spectra exhibited clear interference fringes, showing good qualitative agreement with the simulations.

The measured free spectral range (FSR) and overall spectral response of the devices are consistent with the design predictions, falling within the expected tolerances due to fabrication variations such as waveguide width deviations or slight imperfections in grating couplers. Minor differences in fringe visibility and insertion loss are observed but remain within acceptable limits.

These observations confirm that the fabricated MZIs behave as predicted by the theoretical and simulated models, validating both the design methodology and the fabrication process.

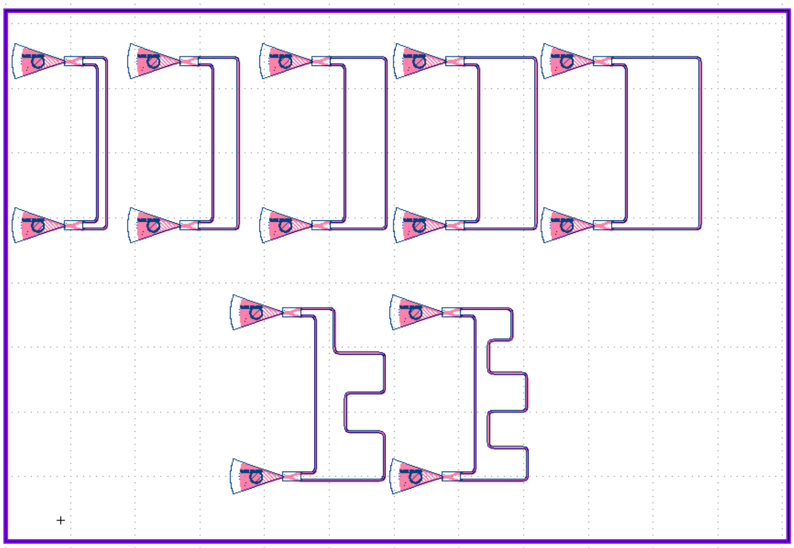


Fig. 5: KLayout design showing the seven Mach–Zehnder interferometers with different ΔL values.

6. Conclusions

In this work, we have presented the design, modeling, and numerical simulation of a Mach–Zehnder interferometer using a silicon-on-insulator (SOI) platform. The study combined theoretical analysis with computational modeling to characterize the device performance. The simulations confirm that the effective and group indices of the waveguide are wavelength dependent, influencing the interferometer's transfer function and free spectral range (FSR). The parameter sweep for different arm length differences demonstrated a clear correlation between ΔL and the FSR, consistent with the theoretical model.